> $eval\left(\frac{1}{2}\right)$

$$\frac{1}{2} \quad (1)$$

> $evalf\left(\frac{1}{2}\right)$

$$0.5000000000 \quad (2)$$

> $eval(\exp(1))$

$$e \quad (3)$$

> $evalf(\exp(1))$

$$2.718281828 \quad (4)$$

> $eval(\mathrm{sqrt}(3))$

$$\sqrt{3} \quad (5)$$

> $evalf(\mathrm{sqrt}(3))$

$$1.732050808 \quad (6)$$

> $eval(\mathrm{Pi})$

$$\pi \quad (7)$$

> $evalf(\mathrm{Pi})$

$$3.141592654 \quad (8)$$

> $eval(\sin(0.1))$

$$0.09983341665 \quad (9)$$

> $evalf(\sin(0.1))$

$$0.09983341665 \quad (10)$$

> $expr := (x^2 + 2\cdot x - 1)^3 \cdot (x^2 - 2)$

$$expr := (x^2 + 2\,x - 1)^3\,(x^2 - 2) \quad (11)$$

> $expand(expr)$

$$x^8 + 6\,x^7 + 7\,x^6 - 16\,x^5 - 27\,x^4 + 14\,x^3 + 17\,x^2 - 12\,x + 2 \quad (12)$$

> $expr := 'expr'$

$$expr := expr \quad (13)$$

> $expand(expr)$

$$expr \quad (14)$$

> $expr := (x + n)^5$

$$expr := (x + n)^5 \quad (15)$$

> $expand(expr)$

$$n^5 + 5\,n^4\,x + 10\,n^3\,x^2 + 10\,n^2\,x^3 + 5\,n\,x^4 + x^5 \quad (16)$$

> $expr := 'expr'$

$$expr := expr \quad (17)$$

> $factor(x^8 - 1)$

$$(x - 1)\,(x + 1)\,(x^2 + 1)\,(x^4 + 1) \quad (18)$$

> $a := factor\left(\frac{2 \cdot x^2}{x^3 - 1}\right)$

$$a := \frac{2x^2}{(x-1)(x^2+x+1)} \quad (19)$$

> $b := factor\left(\frac{3 \cdot x}{x^2 - 1}\right)$

$$b := \frac{3x}{(x-1)(x+1)} \quad (20)$$

> $factor(a+b)$

$$\frac{x(5x^2+5x+3)}{(x-1)(x^2+x+1)(x+1)} \quad (21)$$

> $simplify(\sin^2(x) + \cos^2(x))$

$$1 \quad (22)$$

> $eval(\exp(x) + \ln(x), x = 1)$

$$e \quad (23)$$

> $subs(x = 1, \exp(x) + \ln(x))$

$$e + \ln(1) \quad (24)$$

> $solve(x^2 - 4 \cdot x + 3 = 0, x)$

$$3, 1 \quad (25)$$

> $solve(x^2 \cdot y + 2 \cdot y - x = 0, y)$

$$\frac{x}{x^2+2} \quad (26)$$

> $solve(x^2 \cdot y + 2 \cdot y - x = 0, x)$

$$\frac{1+\sqrt{-8y^2+1}}{2y}, -\frac{-1+\sqrt{-8y^2+1}}{2y} \quad (27)$$

> $fsolve(x - \cos(x) = 0, x)$

$$0.7390851332 \quad (28)$$

> $fsolve(x - \cos(x) = 0, x = -1..1)$

$$0.7390851332 \quad (29)$$

> $fsolve(x^5 - 3 \cdot x^3 - 1 = 0, x = -2..2)$

$$-1.668777593, -0.7418139305, 1.782308780 \quad (30)$$

> $fsolve(\{4 \cdot x + 3 \cdot y = 10, 3 \cdot x - y = 1\}, \{x, y\})$

$$\{x = 1., y = 2.\} \quad (31)$$

> $f := x \mapsto \exp(x) - \sin(x)$

$$f := x \mapsto e^x - \sin(x) \quad (32)$$

> $f(0)$

$$1 \quad (33)$$

> $f(-1)$

$$e^{-1} + \sin(1) \quad (34)$$

```
> D(f)(0)
```

$$0 \quad (35)$$

```
> D(f)(1)
```

$$e - \cos(1) \quad (36)$$

```
> D(f)(x)
```

$$e^{x} - \cos(x) \quad (37)$$

```
> (D@@2)(f)
```

$$x \mapsto e^{x} + \sin(x) \quad (38)$$

```
> diff(f(x), x)
```

$$e^{x} - \cos(x) \quad (39)$$

```
> diff(f(x), x$2)
```

$$e^{x} + \sin(x) \quad (40)$$

```
> int(f(x), x)
```

$$\cos(x) + e^{x} \quad (41)$$

```
> int(f(x), x = -1..1)
```

$$-e^{-1} + e \quad (42)$$

```
> evalf(%)
```

$$2.350402387 \quad (43)$$

```
> f := 'f'
```

$$f := f \quad (44)$$

```
> g := exp(x) - sin(x)
```

$$g := e^{x} - \sin(x) \quad (45)$$

```
> eval(g, x = 0)
```

$$1 \quad (46)$$

```
> d := diff(g, x)
```

$$d := e^{x} - \cos(x) \quad (47)$$

```
> eval(d, x = 0)
```

$$0 \quad (48)$$

```
> int(g, x)
```

$$\cos(x) + e^{x} \quad (49)$$

```
> int(g, x = -1..1)
```

$$-e^{-1} + e \quad (50)$$

```
> evalf(%)
```

$$2.350402387 \quad (51)$$

```
> f := (diff(g, x$2))
```

$$f := e^{x} + \sin(x) \quad (52)$$

```
> eval(f, x = 0)
```

$$1 \quad (53)$$

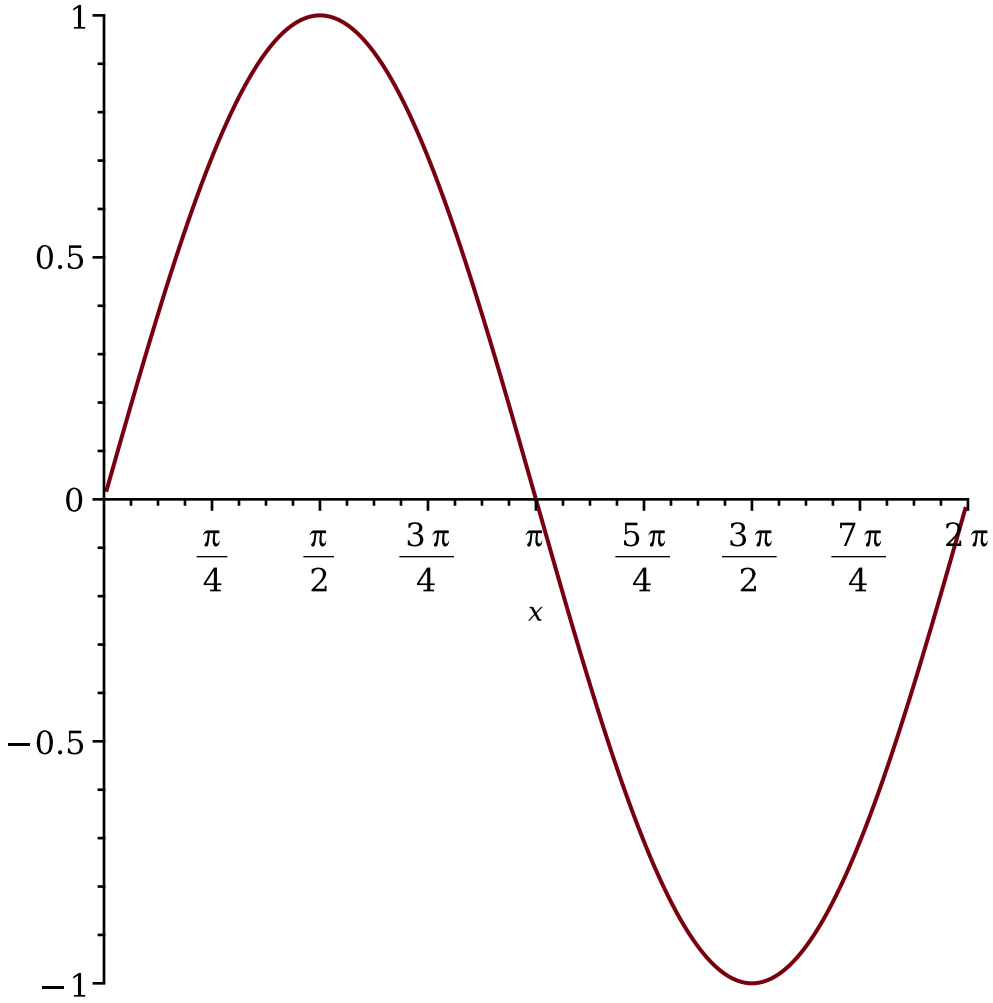
> $limit\left(\frac{\sin(x)}{x}, x = 0\right)$

$$1 \quad (54)$$

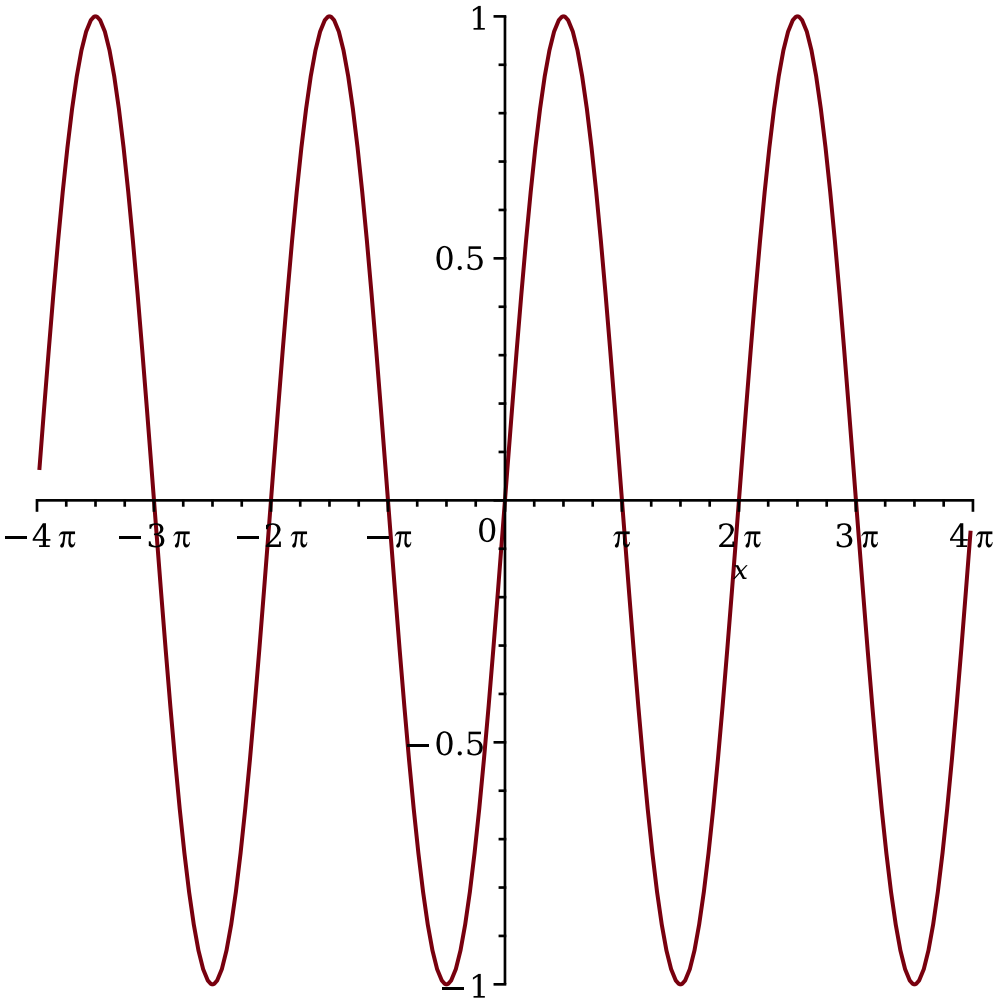
> $limit\left(\frac{(\cos(x)+1)}{x-\mathrm{Pi}}, x = \mathrm{Pi}\right)$

$$0 \quad (55)$$

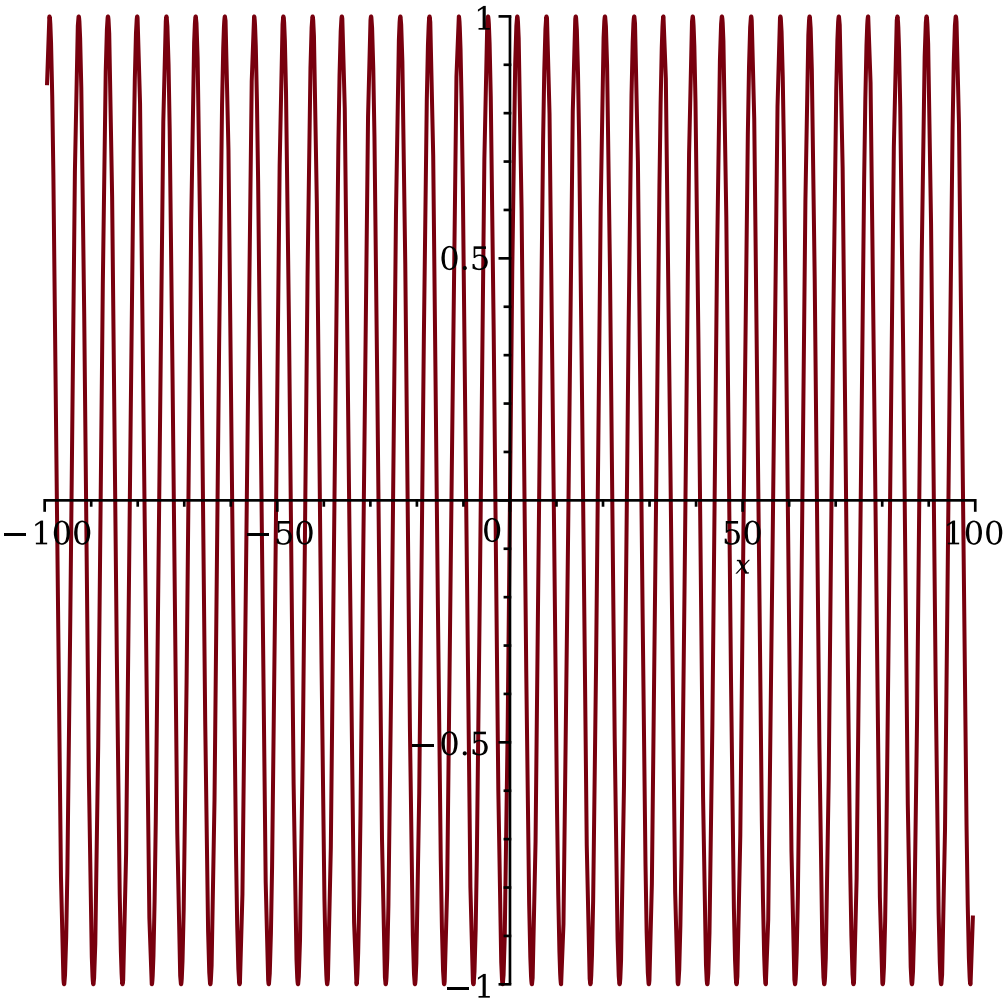
> $plot(\sin(x), x = 0..2\cdot\mathrm{Pi})$



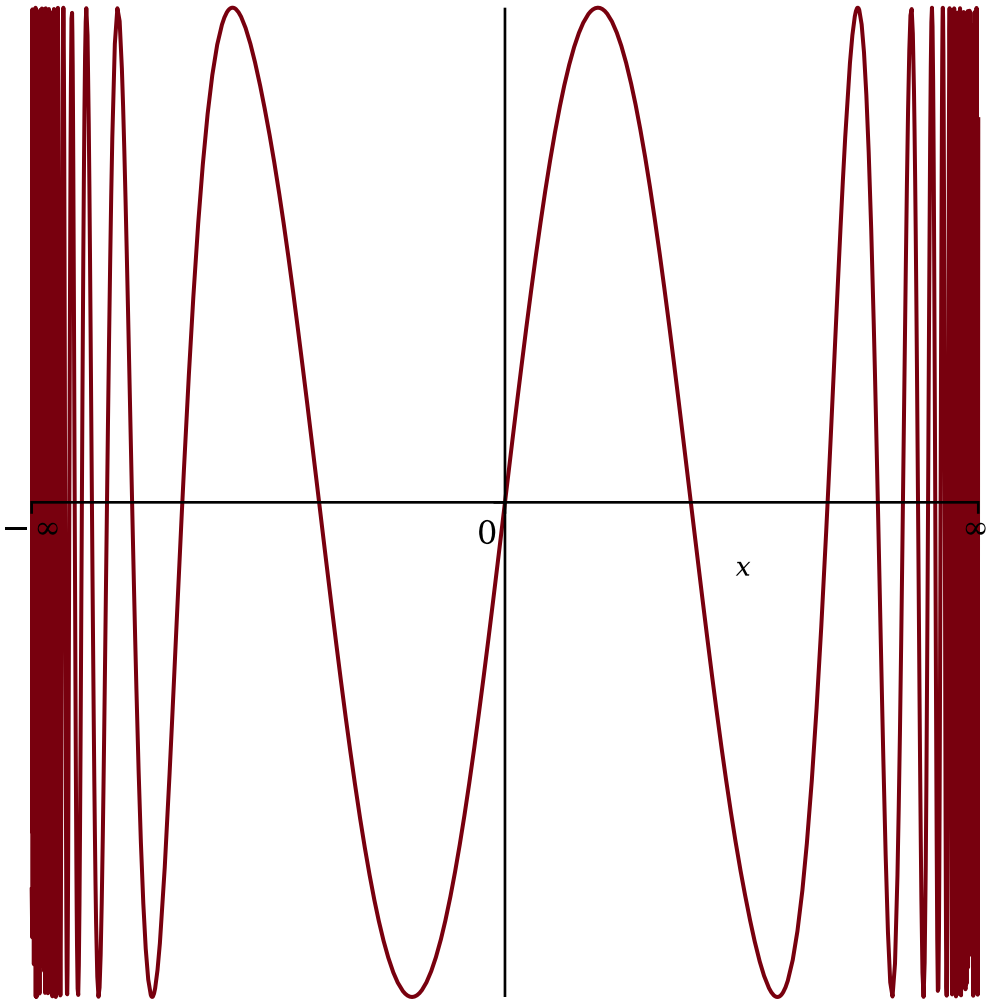
> $plot(\sin(x), x = -4\cdot\mathrm{Pi}..4\cdot\mathrm{Pi})$



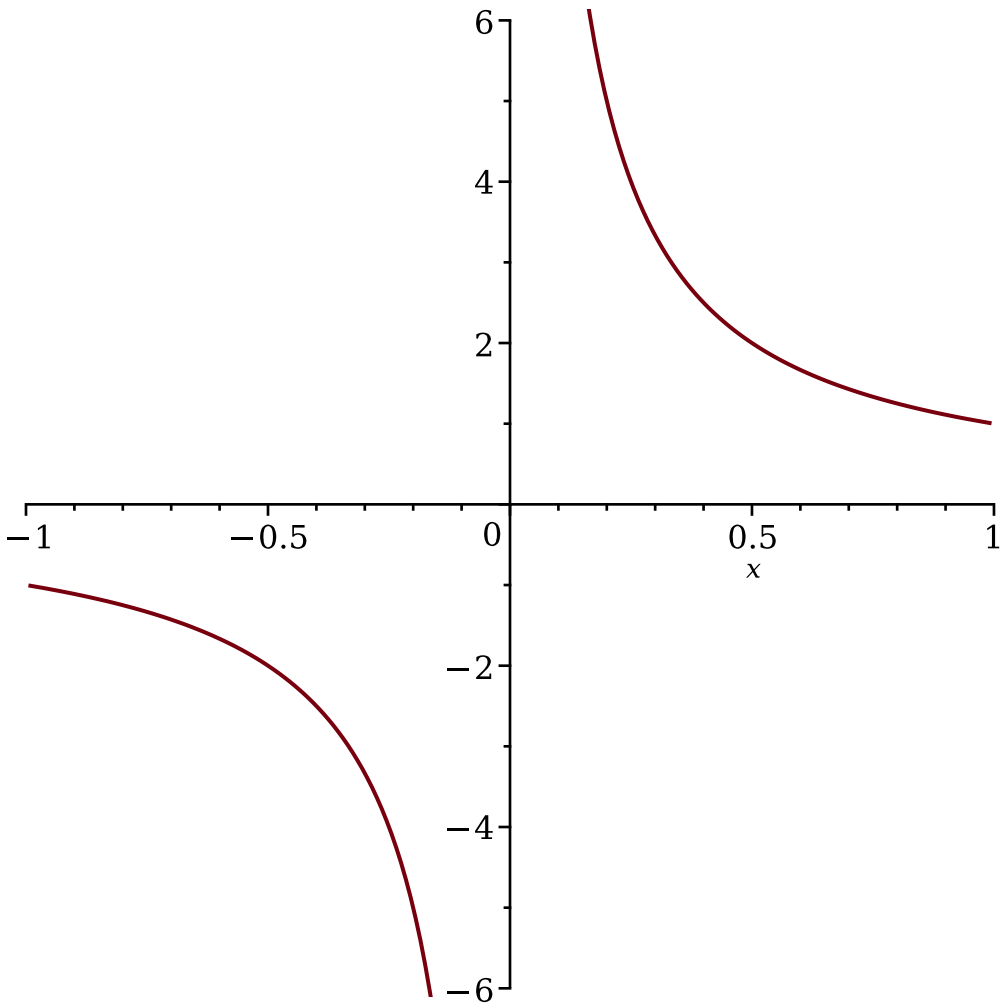

```
> plot(sin(x), x = -100..100)
```



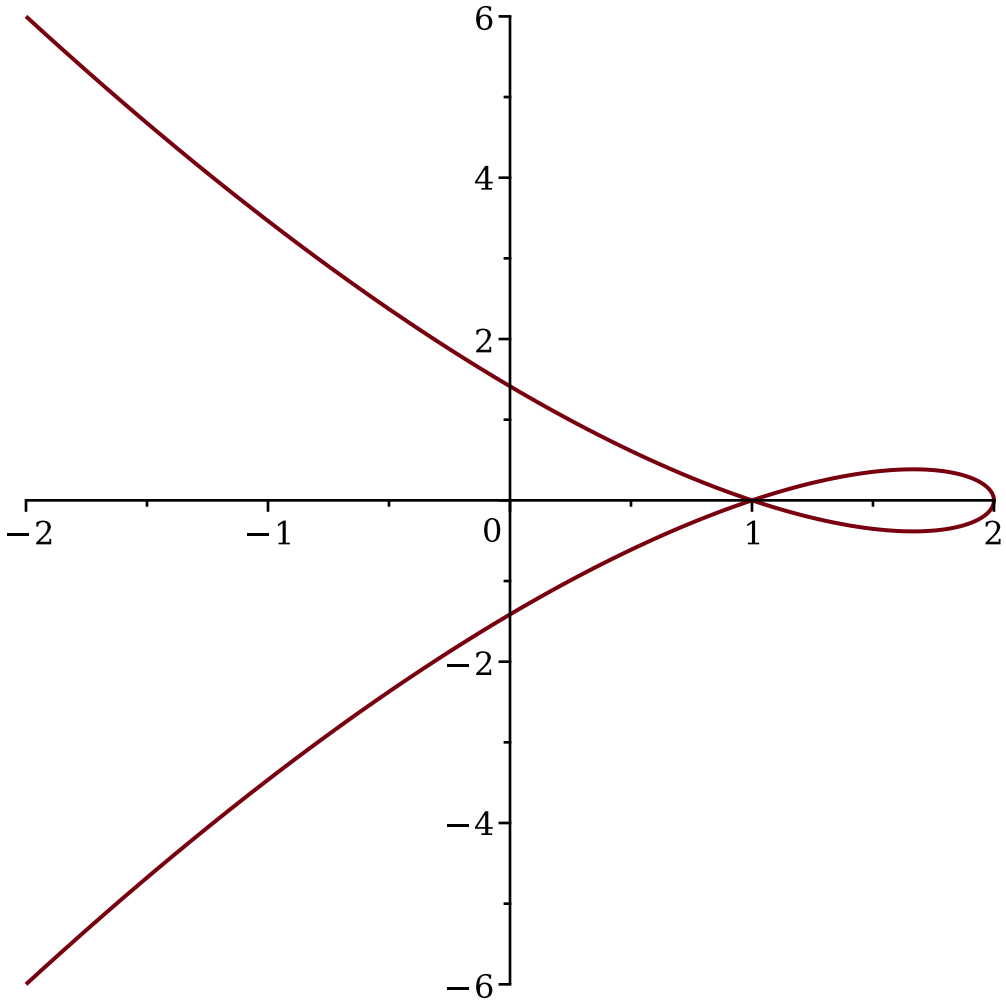
> $plot(\sin(x), x = -\text{infinity}..\text{infinity})$



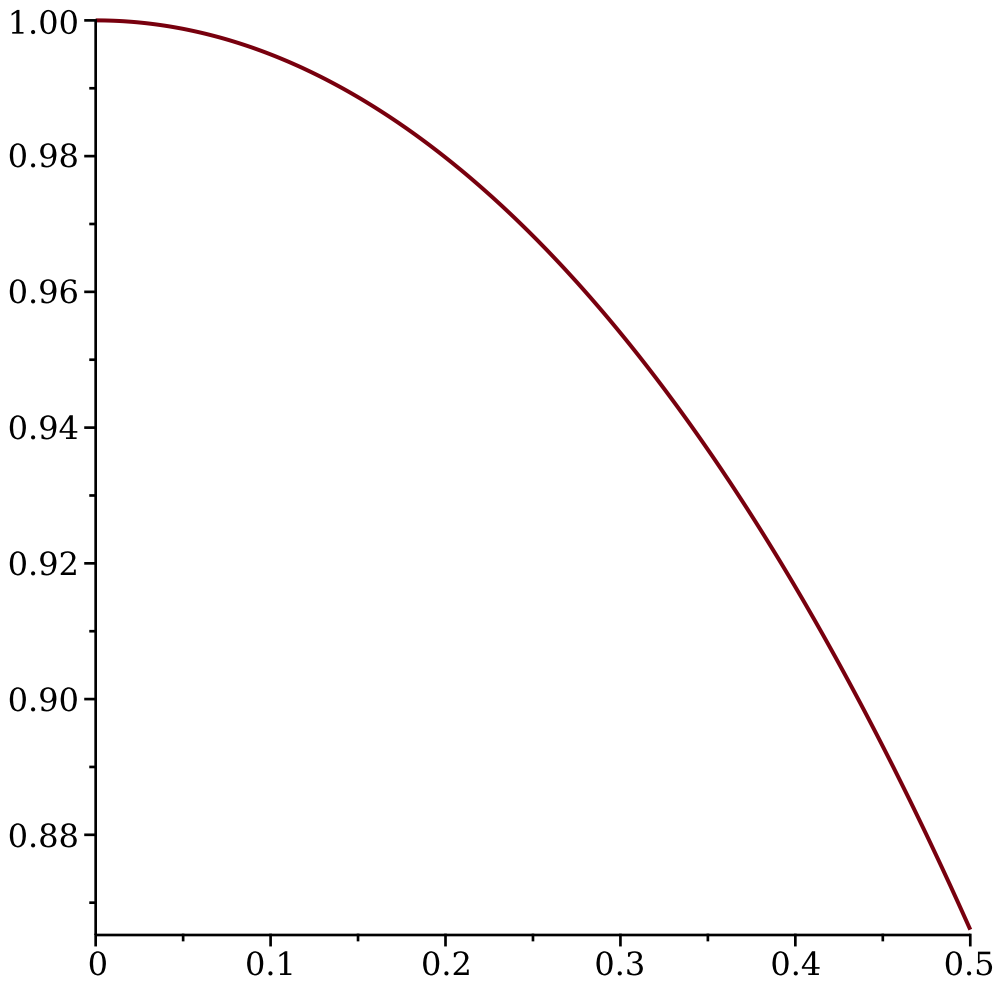
> $plot\left(\frac{1}{x}, x=-1..1\right)$



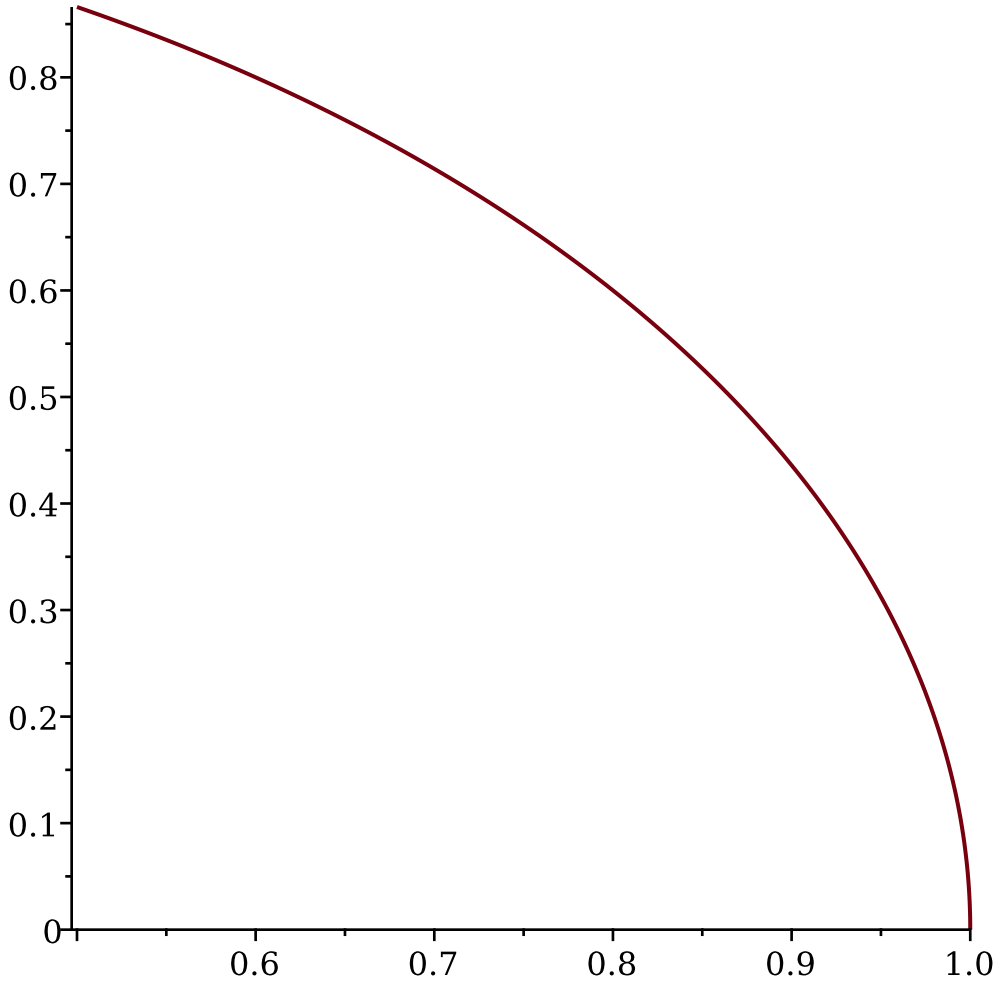

```
> plot([2 - t^2, t - t^3, t = -2..2])
```



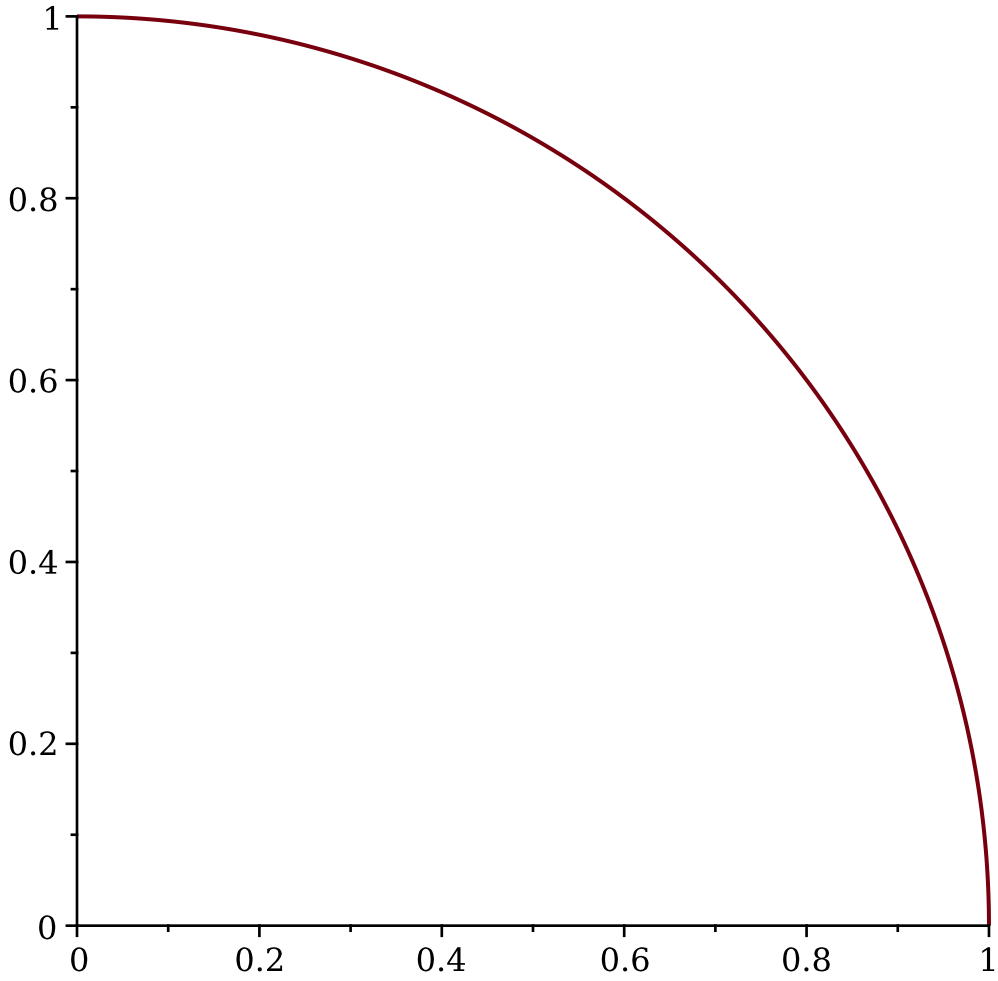
> $plot\left(\left[\sin(x), \cos(x), x = 0 .. \frac{\mathrm{Pi}}{6}\right]\right)$



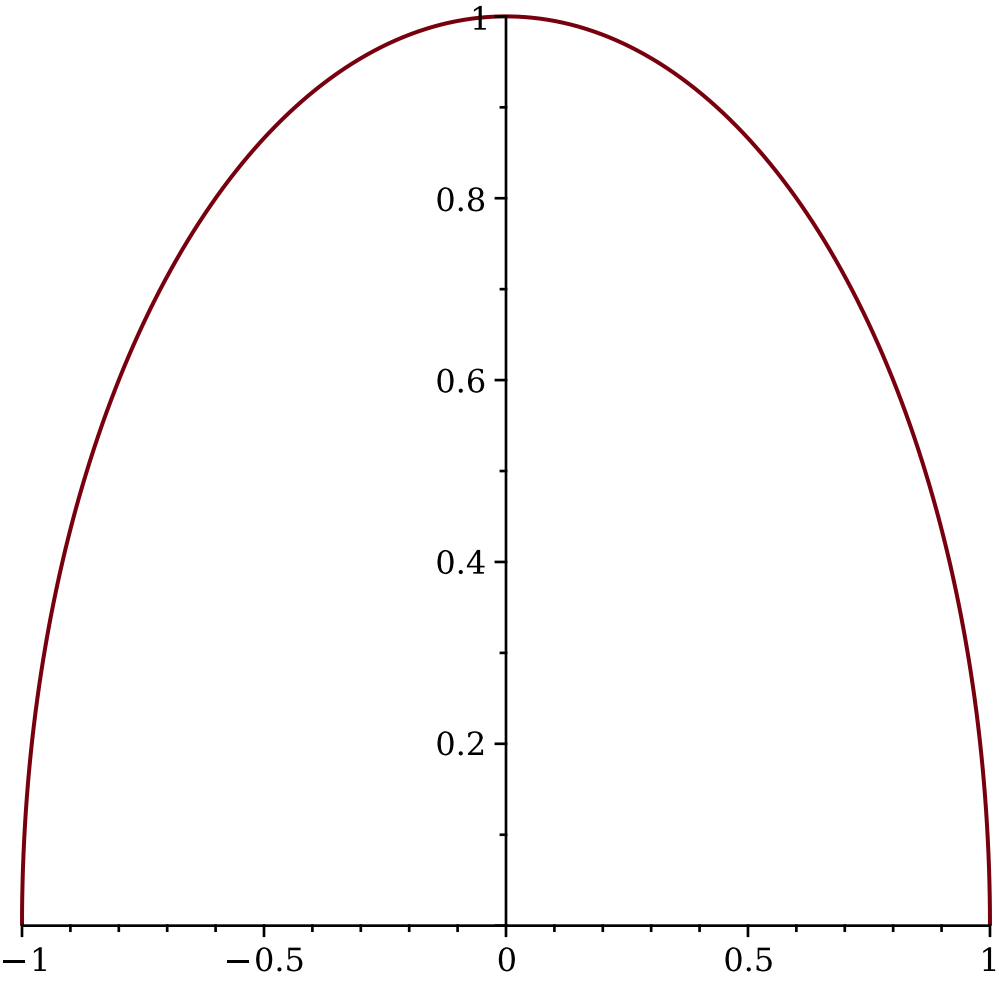
> $plot\left(\left[\cos(t), \sin(t), t = 0 .. \frac{\mathrm{Pi}}{3}\right]\right)$



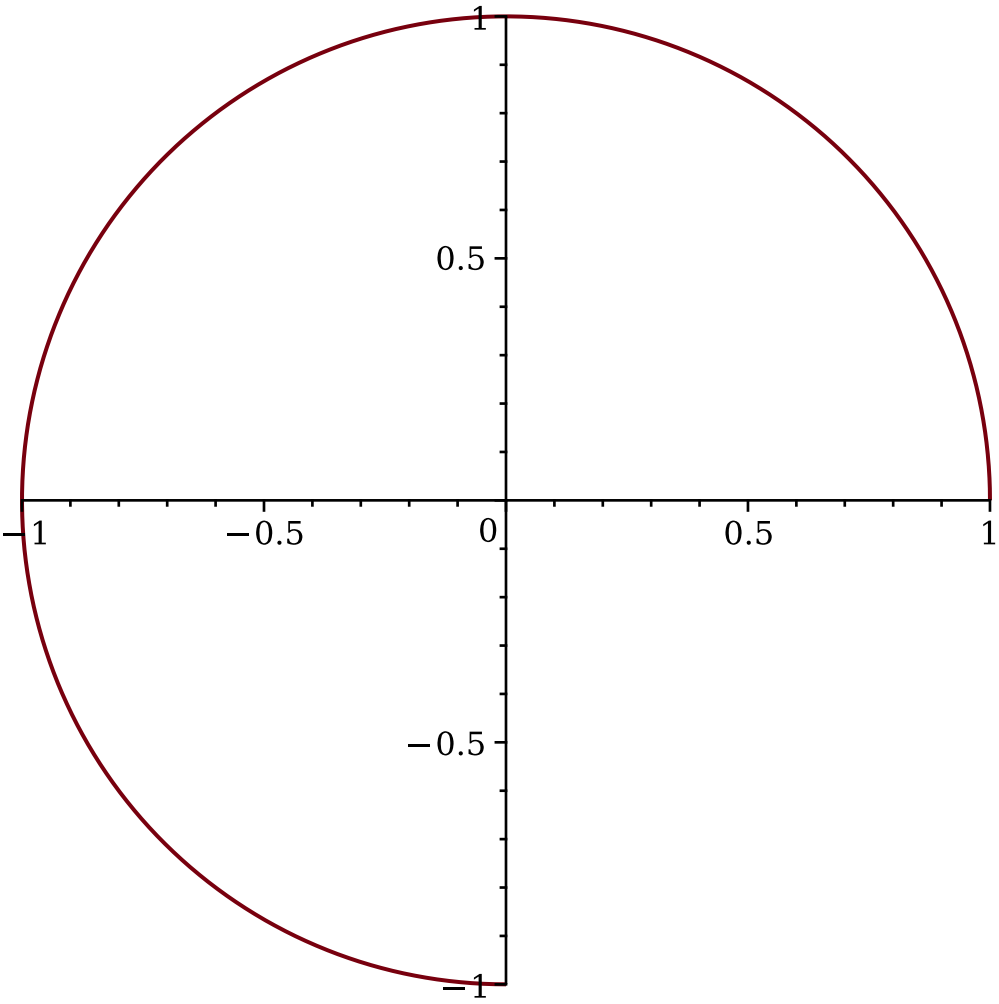

```
> plot([cos(t), sin(t), t = 0 .. Pi/2])
```



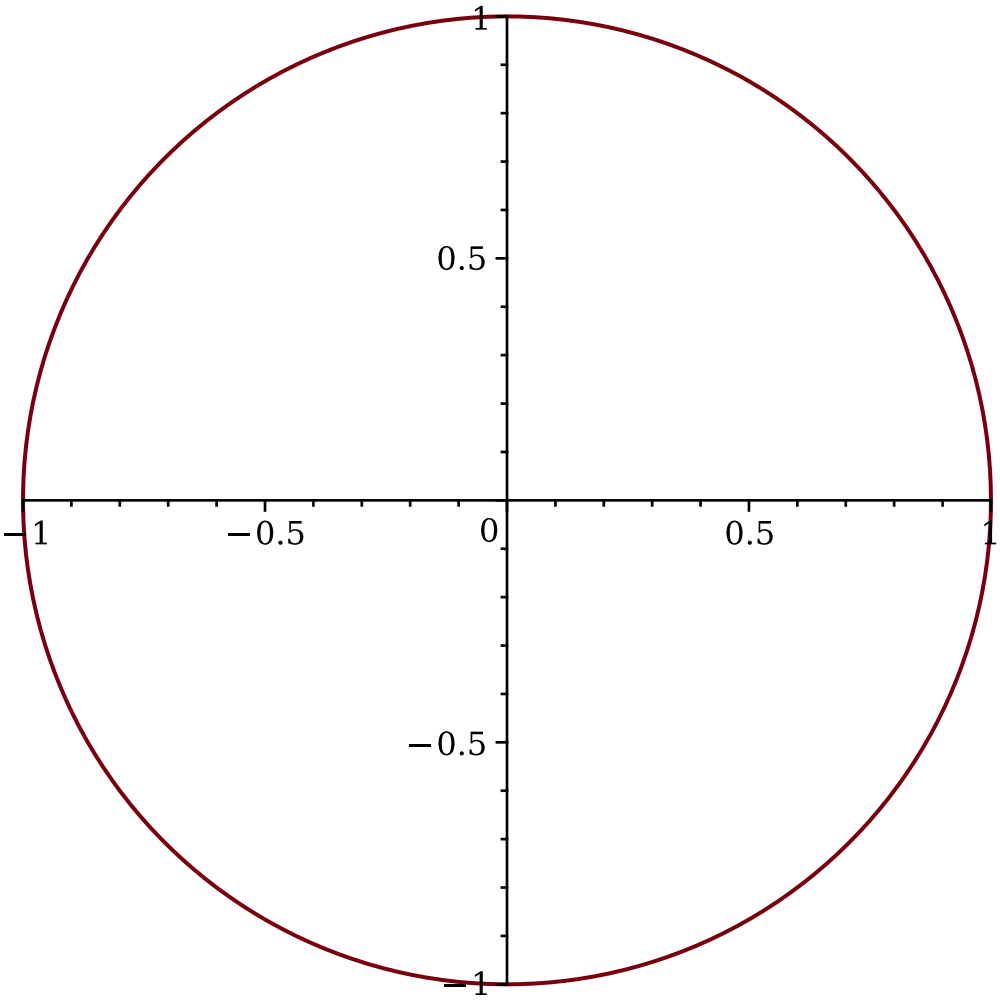
> $plot([\cos(t), \sin(t), t = 0 .. \mathrm{Pi}])$



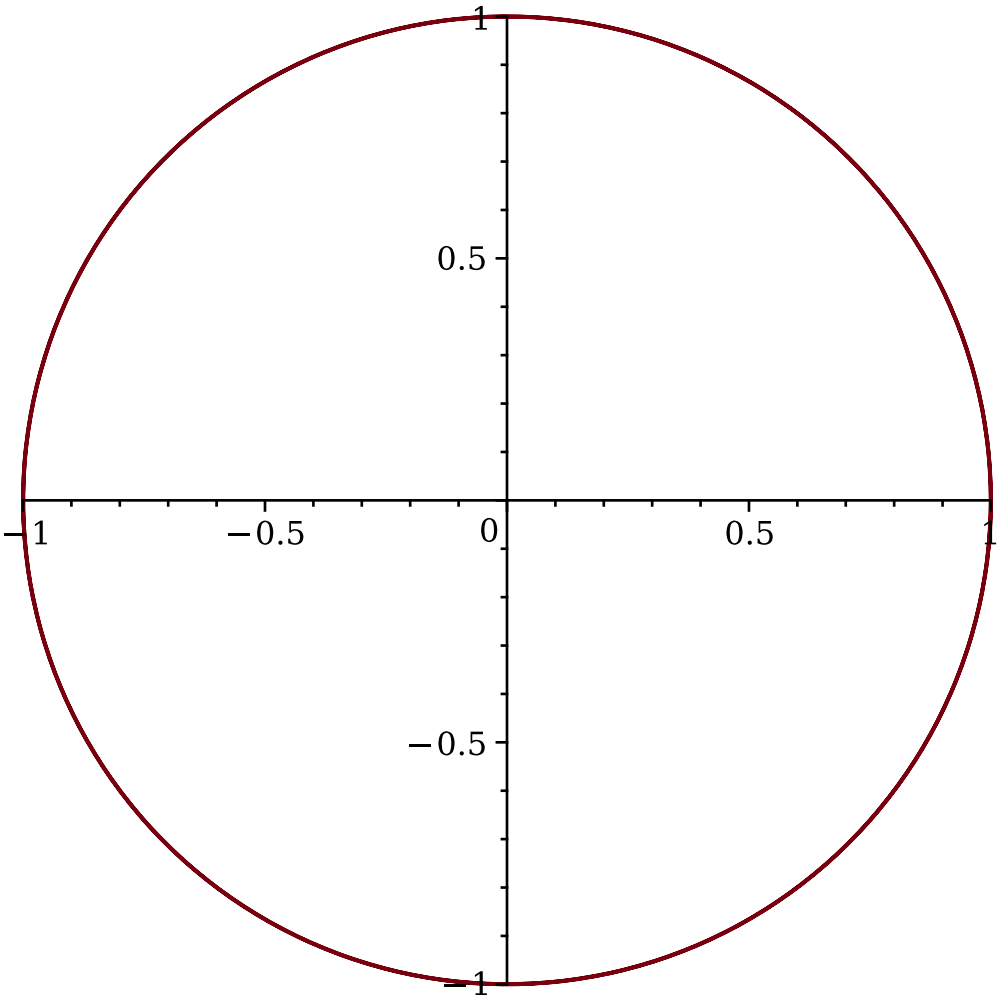
> $plot([\cos(t), \sin(t), t = 0 .. 3*\mathrm{Pi}/2])$



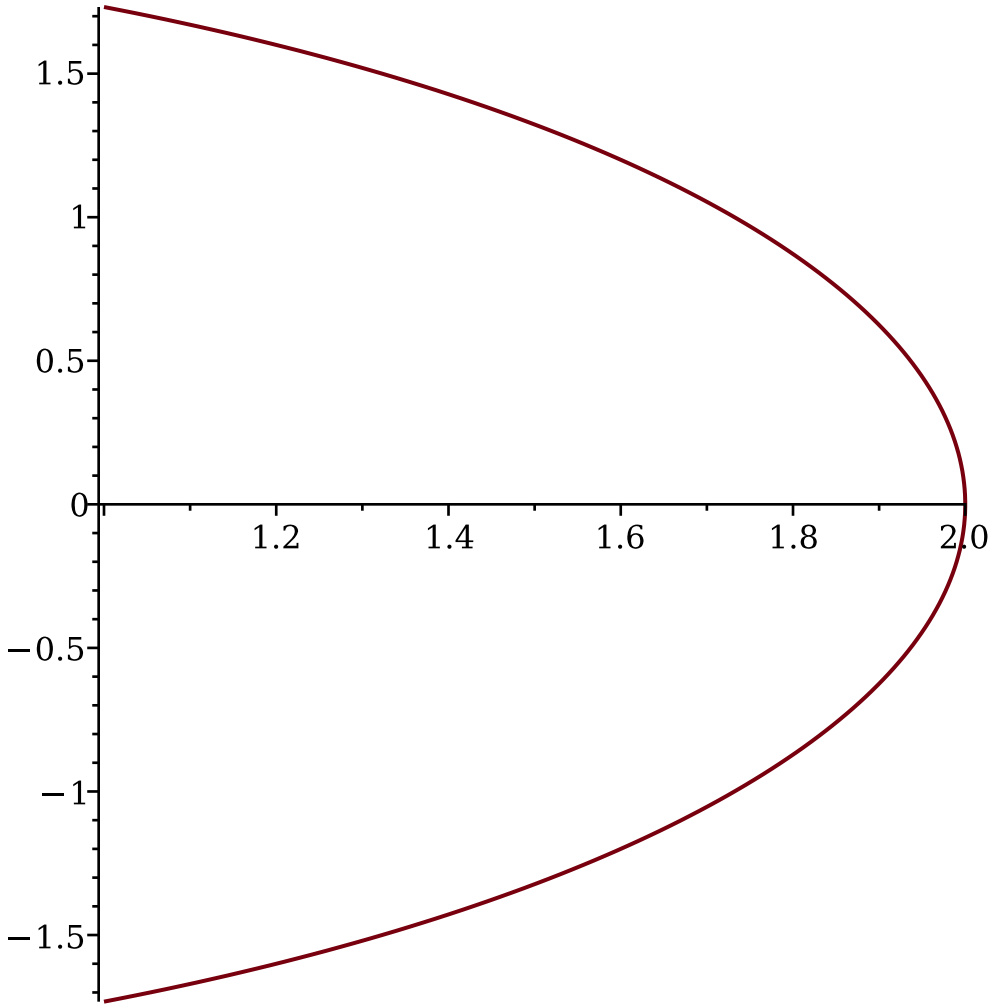
> $plot([\cos(t), \sin(t), t = 0 .. 2*\mathrm{Pi}])$



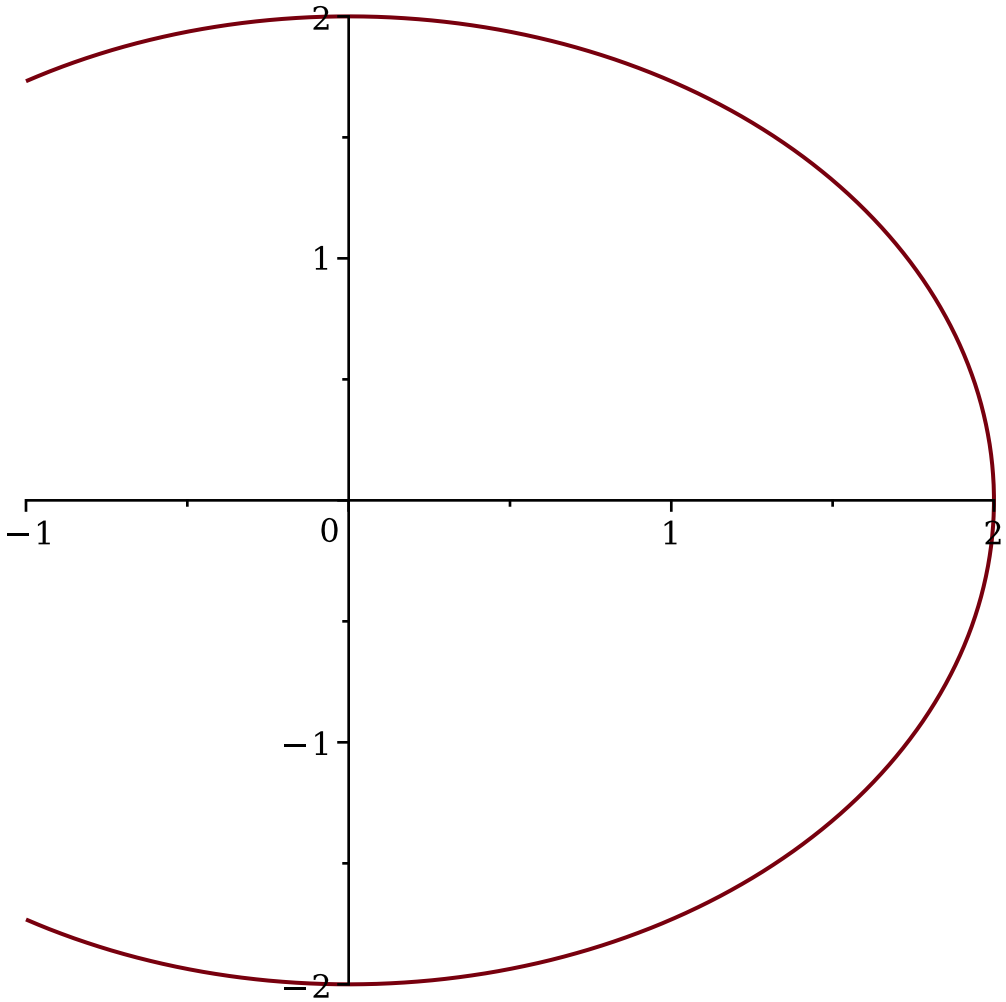
> $plot([\cos(t), \sin(t), t = 0 .. 4*\mathrm{Pi}])$



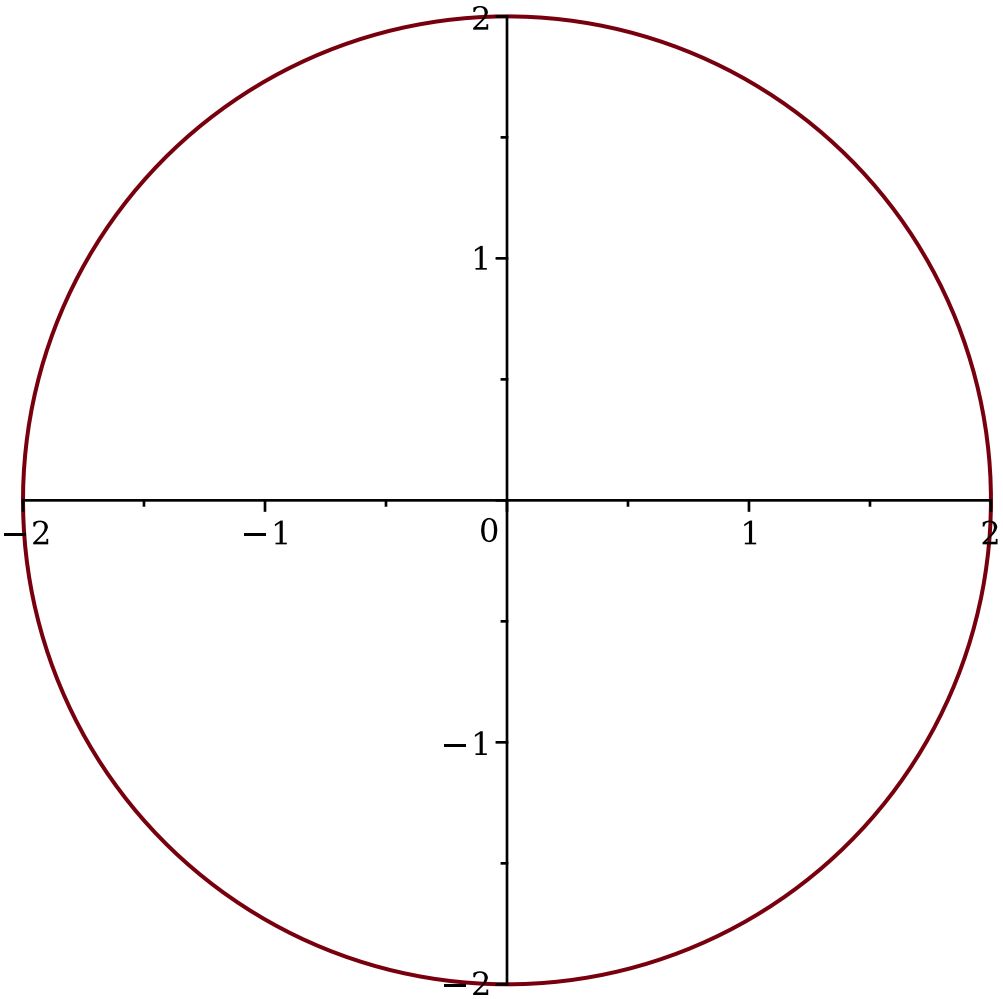
> $plot\left(\left[2\cdot\cos\left(\frac{t}{3}\right), 2\cdot\sin\left(\frac{t}{3}\right), t=-\mathrm{Pi}..\mathrm{Pi}\right]\right)$



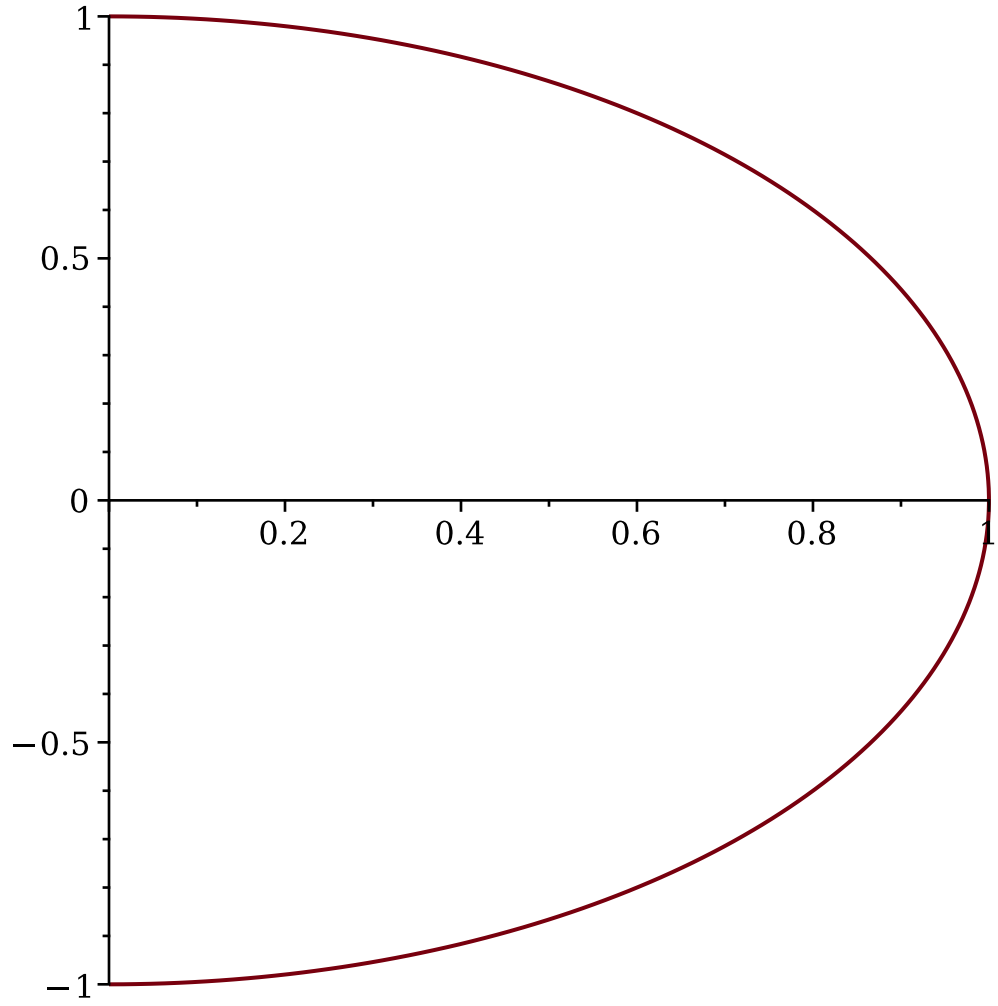
> $plot\left(\left[2\cdot\cos\left(\frac{t}{3}\right), 2\cdot\sin\left(\frac{t}{3}\right), t=-2\cdot \mathrm{Pi}..2\cdot \mathrm{Pi}\right]\right)$



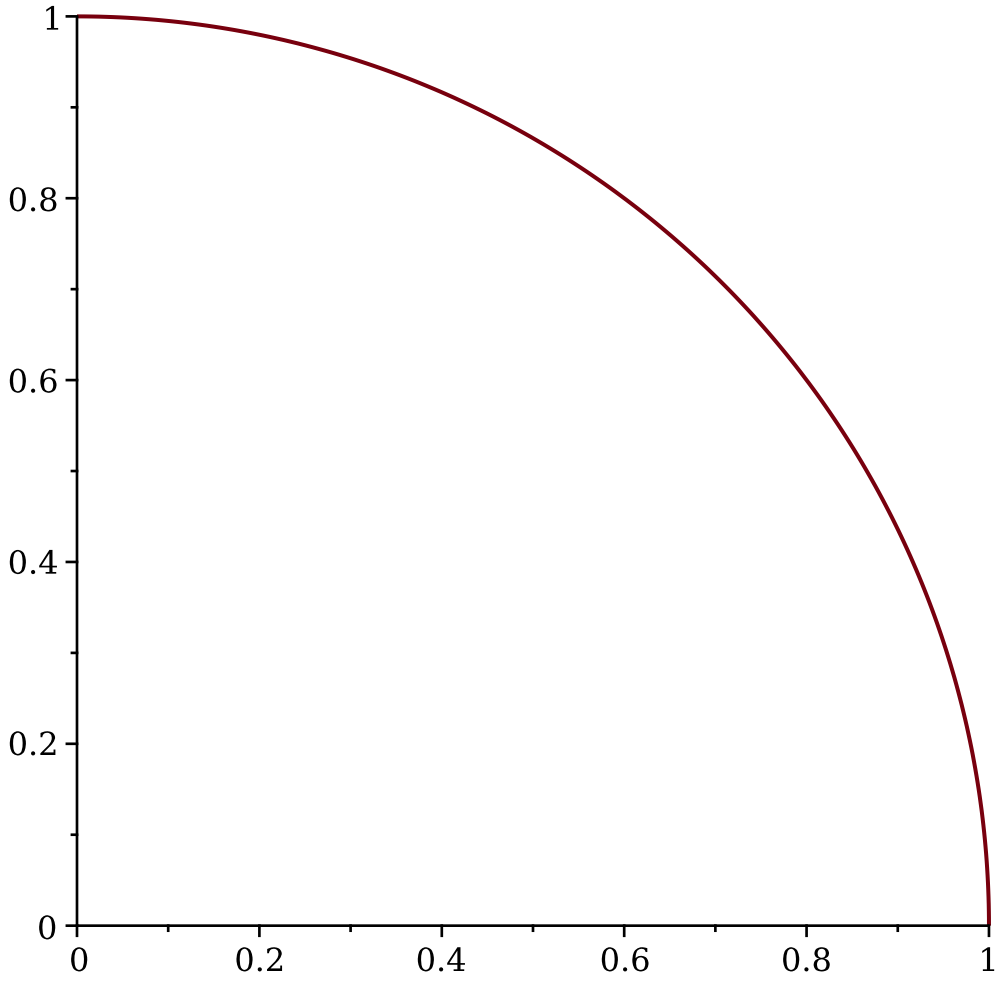
> $plot\left(\left[2\cdot\cos\left(\frac{t}{3}\right), 2\cdot\sin\left(\frac{t}{3}\right), t=-3\cdot \mathrm{Pi}..3\cdot \mathrm{Pi}\right]\right)$



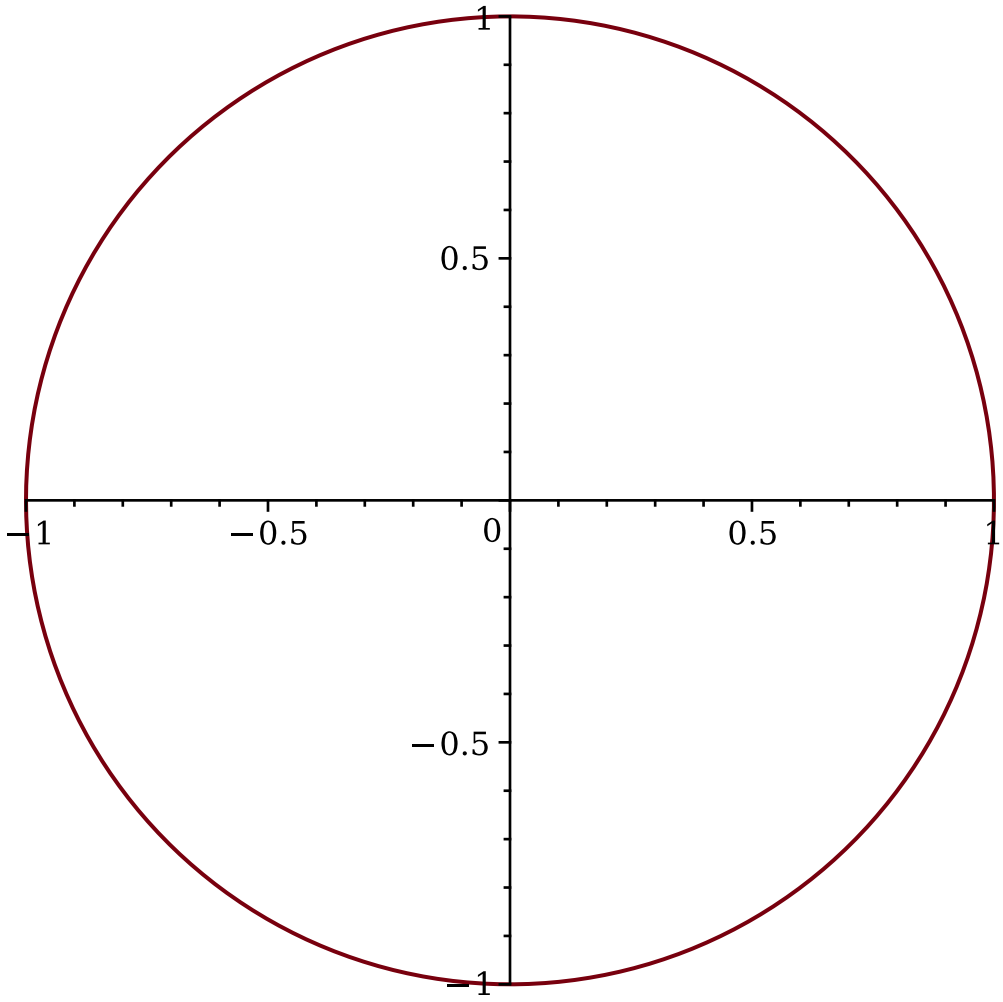
> $plot\left(\left[\cos(4\cdot t), \sin(4\cdot t), t = -\frac{\mathrm{Pi}}{8} .. \frac{\mathrm{Pi}}{8}\right]\right)$



> $plot\left(\left[\cos(4\cdot t), \sin(4\cdot t), t = 0..\frac{\mathrm{Pi}}{8}\right]\right)$



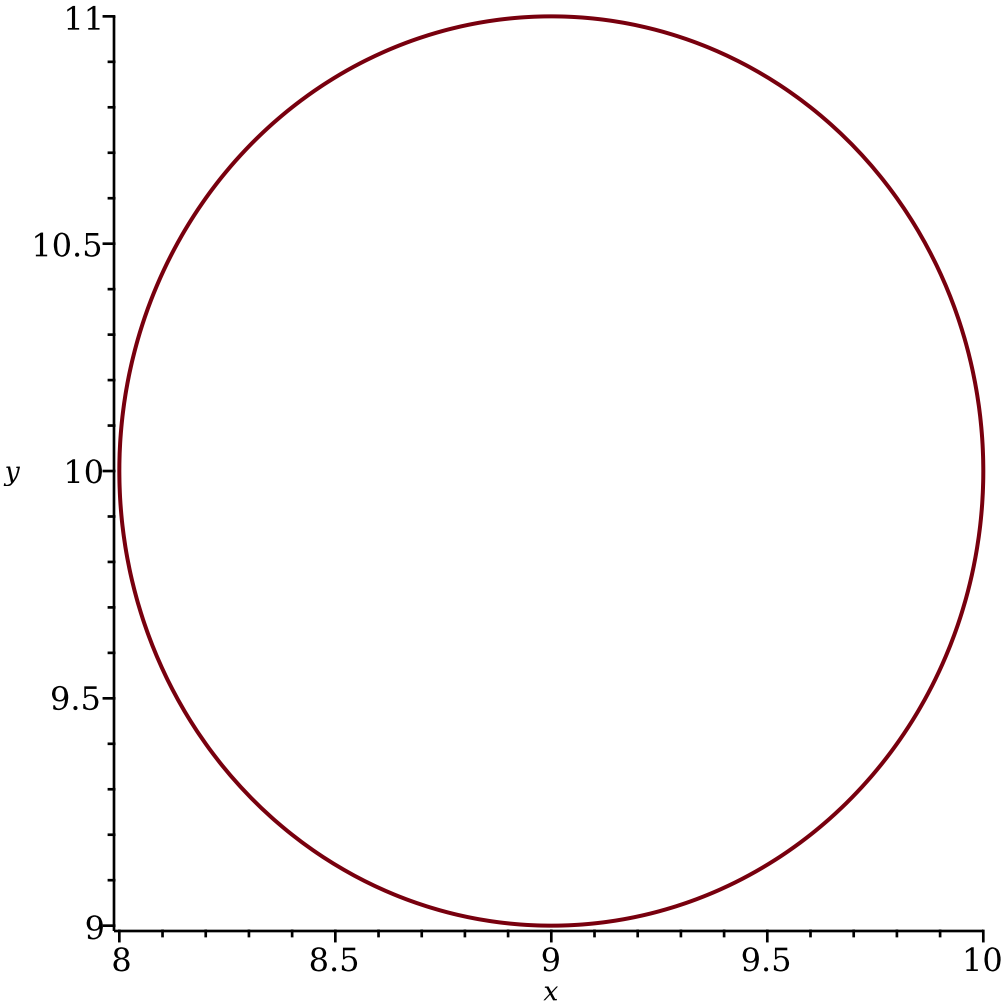
> $plot\left(\left[\cos(4\cdot t), \sin(4\cdot t), t = -\frac{\mathrm{Pi}}{4} .. \frac{\mathrm{Pi}}{4}\right]\right)$



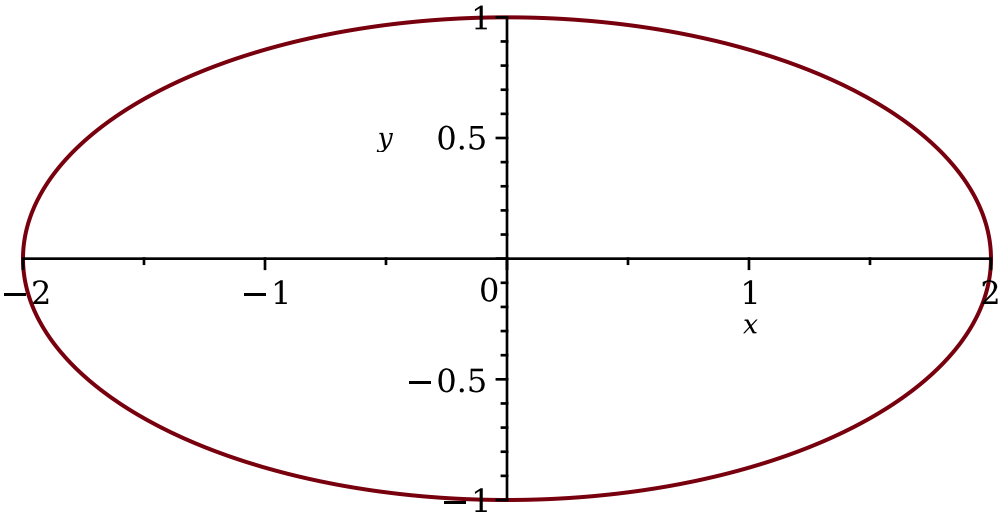

```
> with(plots)
```

[*animate, animate3d, animatecurve, arrow, changecoords, complexplot, complexplot3d, conformal, conformal3d, contourplot, contourplot3d, coordplot, coordplot3d, densityplot, display, dualaxisplot, fieldplot, fieldplot3d, gradplot, gradplot3d, implicitplot, implicitplot3d, inequal, interactive, interactiveparams, intersectplot, listcontplot, listcontplot3d, listdensityplot, listplot, listplot3d, loglogplot, logplot, matrixplot, multiple, odeplot, pareto, plotcompare, pointplot, pointplot3d, polarplot, polygonplot, polygonplot3d, polyhedra_supported, polyhedraplot, rootlocus, semilogplot, setcolors, setoptions, setoptions3d, shadebetween, spacecurve, sparsematrixplot, surfdata, textplot, textplot3d, tubeplot*] **(56)**

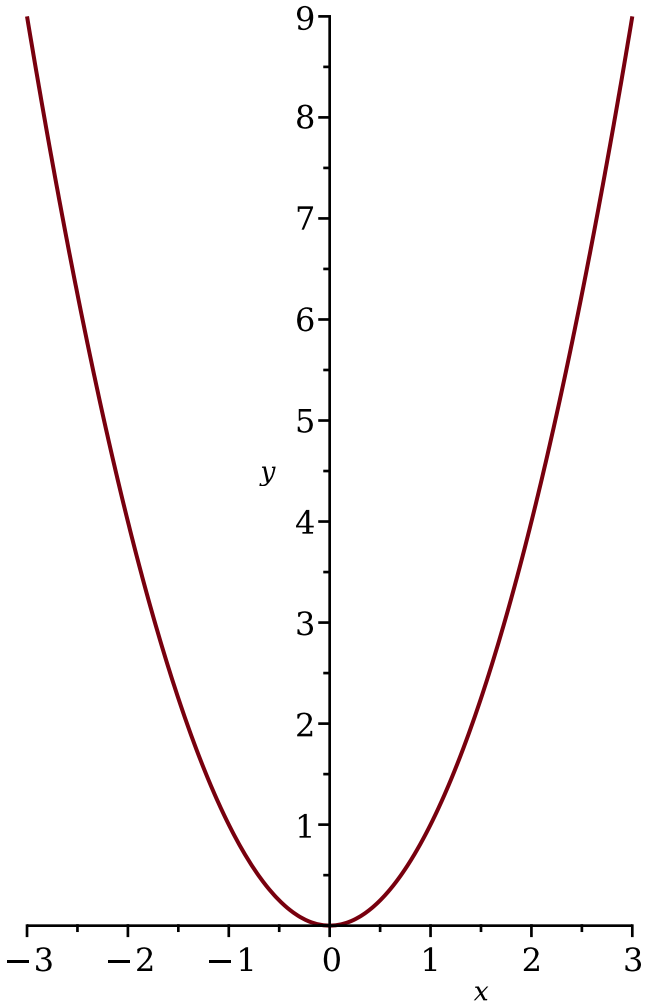
```
> implicitplot( (x-9)^2 + (y-10)^2 = 1, x=-20..20, y=-40..40)
```



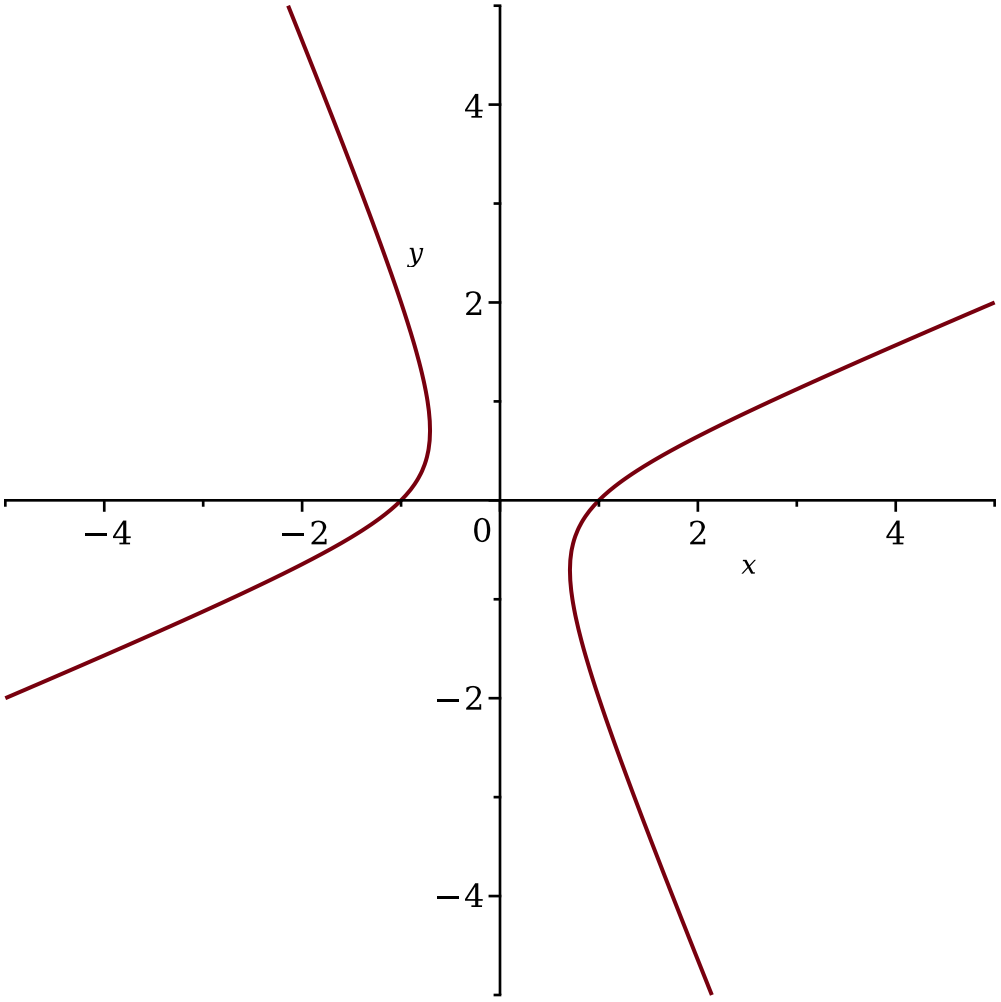

```
> implicitplot(x^2/4 + y^2 = 1, x = -3 .. 3, y = -2 .. 2, scaling = constrained)
```



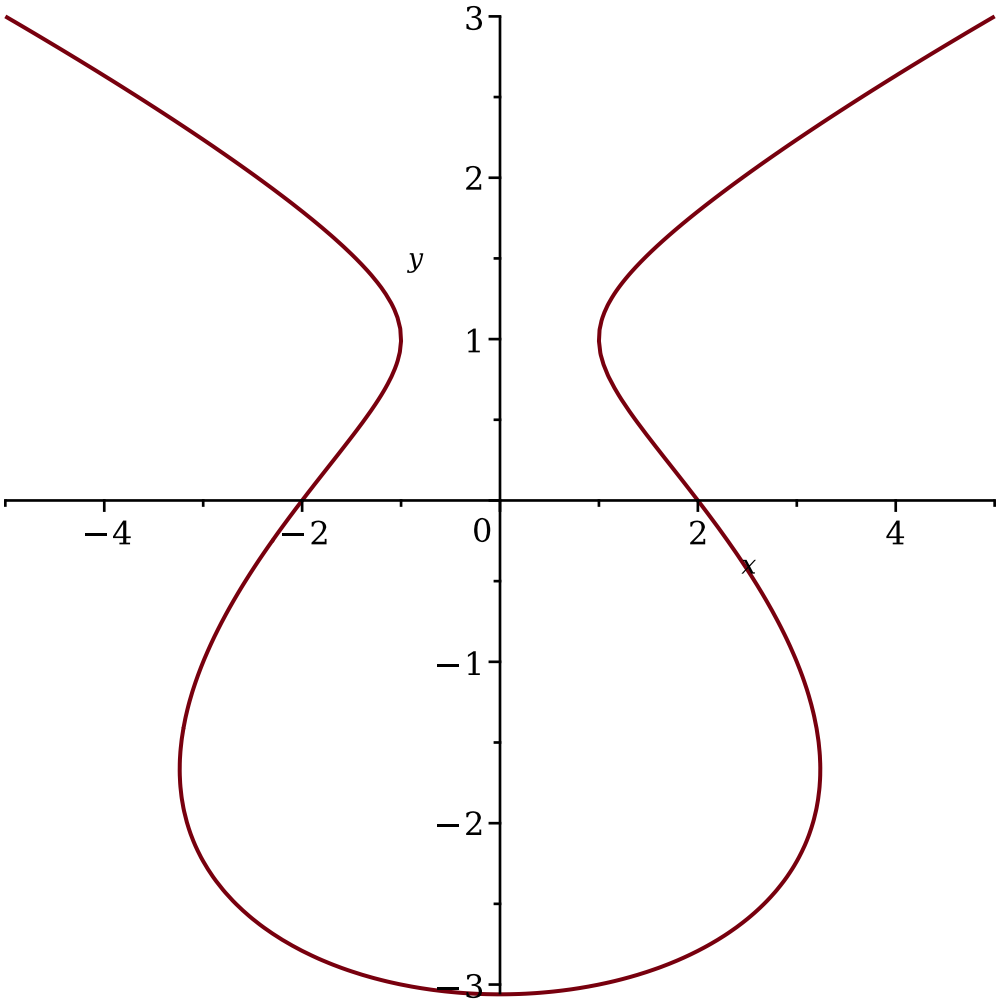
> $implicitplot(y = x\hat{}2, x = -3 .. 3, y = 0 .. 10, scaling = constrained)$



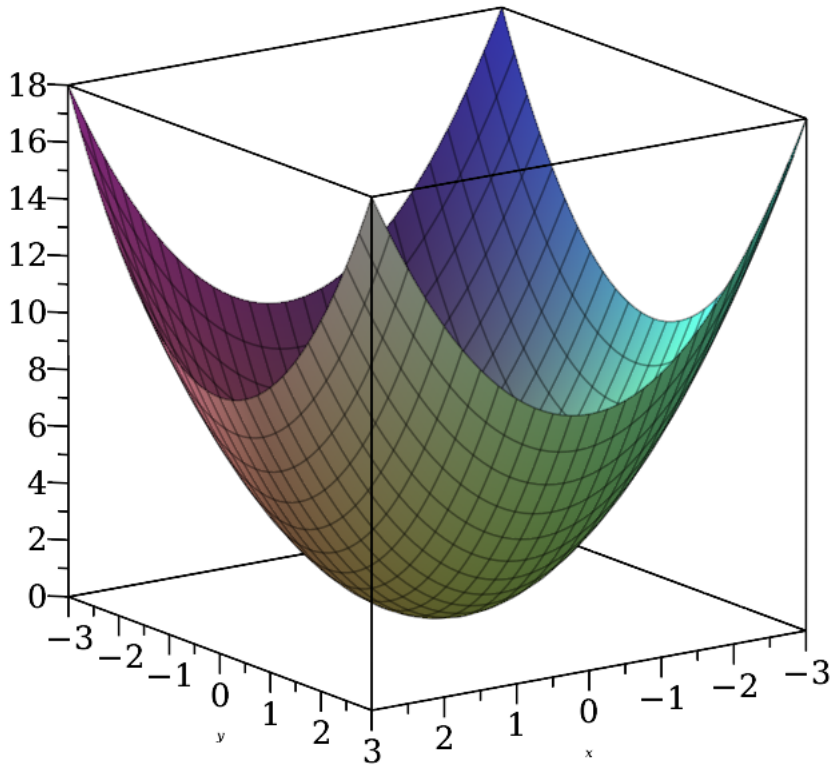
> $implicitplot(x^2 - 2 \cdot x \cdot y - y^2 = 1, x = -5..5, y = -5..5)$



> $implicitplot(y^3 + y^2 - 5 \cdot y - x^2 = -4, x = -5..5, y = -5..5)$



> $plot3d(x^2+y^2, x=-3..3, y=-3..3)$


18
16
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3
y
2
1
0
−1
−2
−3
x